

BEYOND ONSET DETECTION: UNSUPERVISED NOTE BOUNDARY DETECTION FOR BOWED STRINGS

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ABSTRACT

Note boundary detection in bowed string performance is a fine-grained audio analysis task that has received little systematic attention. Unlike onset detection, which targets the beginning of a note, note boundary detection aims to identify transitions between adjacent notes, which in bowed string playing manifest as broadband spectral changes rather than sharp energy transients. We conduct a comparative evaluation of three unsupervised boundary strength functions based on log-mel spectrogram frame differences, originally proposed as the boundary proposal stage of a bow-change detection system, and compare them against five signal-processing onset detection baselines. On a dataset of 9 recordings spanning viola, violin, and cello, our best variant achieves a tonal mean F1 of 77 % at ± 100 ms without any training, outperforming all baselines including librosa onset detection (71 %). The proposed variants retain their advantage across tolerance windows from ± 50 ms to ± 150 ms, suggesting that spectral shape change can provide a more effective cue for temporally localising bowed-string note boundaries than the evaluated onset detection baselines.

1. INTRODUCTION

Onset detection is one of the most studied tasks in music information retrieval [1, 2], supporting applications from automatic music transcription [3] to performance analysis [4]. However, for bowed string instruments, a note *onset*, usually understood as a sharp increase in energy, often poorly characterises the acoustic reality. Bowed string notes may begin gradually, while the perceptually salient event is frequently the *transition between notes*: a brief region of spectral change caused by bow direction changes, string crossings, or left-hand finger articulation.

We define *note boundary detection* as identifying the onset of the spectral transition between adjacent notes in a monophonic bowed string recording. Unlike onset detection, the target is a transition rather than a beginning; the boundary region may span 100–300 ms in legato playing;

and the relevant cue is spectral shape change across mel bands rather than wideband energy increase.

Prior work on string instrument audio analysis has focused on onset detection [5, 6], playing technique classification [11], and automatic transcription [7]. Note boundary detection as a distinct task has received little systematic attention. We evaluate three unsupervised boundary strength functions from the Stage 1 proposal component of the open-source BowDET toolkit [8] as standalone note boundary detectors. The present study applies them to a broader task formulation—note boundaries rather than bow changes alone—using a distinct evaluation protocol and a separate repertoire subset, and compares them against five signal-processing onset baselines.

2. METHOD

2.1 Boundary Strength Functions

Audio is downsampled to 22,050 Hz. We compute a log-mel spectrogram with 64 mel bands, 2048-point FFT, and 512-sample hop (≈ 23 ms per frame), over a frequency range of 100–8000 Hz.

We evaluate three variants of the same boundary strength idea, differing only in temporal context:

V1 (adjacent frame difference): boundary strength at frame t is the ℓ_2 norm of the difference between adjacent mel vectors:

$$S_{V1}(t) = \|\mathbf{m}_t - \mathbf{m}_{t-1}\|_2 \quad (1)$$

$S_{V1}(t)$ is smoothed with a 5-frame mean filter and min-max normalised.

V2 (left–right window): compares mean mel vectors over adjacent windows of width $w = 5$ frames:

$$S_{V2}(t) = \|\bar{\mathbf{m}}_{[t, t+w)} - \bar{\mathbf{m}}_{[t-w, t)}\|_2 \quad (2)$$

V3 (multi-scale): averages normalised V2 scores over three window sizes $w \in \{3, 5, 8\}$ frames.

For each variant, boundary candidates are peaks of the corresponding boundary strength function above its 75th percentile, with a minimum inter-peak distance of 0.12 s.

2.2 Baselines

We compare against five signal-processing onset detection baselines: **Librosa Onset** (onset_detect, default parameters [9]), which has its own internal peak-picking; **Mel Spectral Flux** (librosa onset_strength



computed from a 64-band mel spectrogram spanning 100–8000 Hz); **HFC**; **Complex Domain** (phase-deviation onset function [10]); and **RMS Diff**. The latter four are peak-picked using the same 75th-percentile threshold and 0.12 s minimum inter-peak distance as the proposed methods.

2.3 Annotation Protocol

Note boundaries were annotated by a single trained viola player using Audacity with visual reference to the log-mel spectrogram, assisted by an open-source annotation tool [8]. Labels mark the onset of the visible spectral transition. Bowed-string note boundaries are perceptually ambiguous and often visually clearer than aurally discrete, requiring instrument-specific expertise to annotate reliably.

3. DATASET

Table 1 summarises the evaluation dataset: 9 recordings from 9 performers across three bowed string instruments, with 1,531 annotated note boundaries.

ID	Instrument	Repertoire	Events
A	Viola	Schumann (Romantic)	176
B	Viola	Paganini (Virtuosic)	360
C	Viola	Biber (Baroque)	98
D	Viola	Penderecki (Contemporary)	19
E	Viola	Reger (Late Romantic)	164
F	Cello	Bach Suite (Baroque)	241
G	Violin	Bach Partita (Baroque)	284
H	Cello	Popular (rock cover)	103
I	Violin	Partita (Baroque)	86
Total			1,531

Table 1. Dataset. D and G excluded from tonal analysis.

We define *tonal repertoire* as recordings using standard bowing technique on pitched notes, excluding extended techniques and passages containing successive note boundaries less than 0.12 s apart. Under this definition, recording D (extended techniques) and recording G (note density violating the 0.12 s inter-peak constraint) are excluded from tonal analysis.

4. RESULTS

Table 2 reports per-recording F1 at the primary ± 100 ms tolerance, while Table 3 evaluates robustness across different temporal windows and exploratory one-to-one IoU-based matching. For the IoU evaluation, each predicted and reference boundary was converted into a 500 ms pseudo-region centred on its timestamp, and a match was counted as correct when its intersection-over-union met or exceeded the specified threshold. The proposed-method rankings remain stable across temporal tolerances, with limited improvement beyond ± 100 ms.

V3 achieves the highest tonal mean F1 of 77 %, followed by V2 (75 %) and V1 (74 %). All three proposed variants outperform all five baselines; librosa onset detection, the strongest baseline, reaches 71 % on tonal recordings.

Method	A	B	C	D	E	F	G	H	I	All	Tonal
V1	66	93	72	35	88	74	44	76	53	67	74
V2	69	84	78	42	89	80	53	65	61	69	75
V3	73	85	80	40	88	82	54	74	60	71	77
Lib. Onset	54	86	74	18	66	87	87	81	48	67	71
Spec. Flux	38	72	52	15	61	82	67	80	66	59	64
HFC	35	48	51	17	46	78	50	81	57	51	57
Complex	43	74	47	31	52	80	43	88	60	58	63
RMS Diff	42	59	50	25	58	73	55	70	53	54	58

Table 2. F1 (%) at ± 100 ms. Tonal excludes D and G.

Method	± 50	± 100	± 150	IoU@0.1	IoU@0.3
V1	64	74	75	75	75
V2	67	75	77	79	79
V3	70	77	79	81	80
Lib. Onset	53	71	71	73	73
Spec. Flux	36	64	66	70	69
HFC	47	57	63	70	67
Complex	55	63	64	68	65
RMS Diff	44	58	61	69	66

Table 3. Tonal mean F1 (%) across temporal tolerances and exploratory pseudo-region IoU evaluation. The proposed variants retain the three highest scores across point- and pseudo-region-based metrics.

Performance varies substantially across repertoire. Paganini (B) and Reger (E) yield high F1 values, consistent with their relatively clear articulations and sharp spectral transitions. Penderecki (D) contains extended techniques that generate dense spurious boundary signals; Bach Partita (G) violates the 0.12 s inter-peak constraint.

Librosa Onset achieves higher F1 than the proposed methods on Bach (F), Bach Partita (G), and Popular (H), consistent with more salient note attacks in these recordings, suggesting the proposed methods may instead be advantageous in legato and expressive passages where spectral changes remain visible despite weak energy transients.

The proposed variants retain their advantage across all tolerance windows and IoU thresholds.

5. DISCUSSION

Mel spectrogram boundary strength functions transfer effectively from bow-change to note-boundary detection, outperforming all five onset baselines on tonal repertoire across the evaluated temporal tolerances and exploratory pseudo-region metrics.

Failure modes include extended techniques and dense passages containing note boundaries less than 0.12 s apart: a supervised Stage 2 classifier [8] may reject spurious candidates from extended techniques, but dense passages instead require a shorter or adaptive inter-peak constraint. Limitations include the small dataset, single-annotator labels, and point rather than interval annotation. Future work will add multi-annotator validation and true interval-overlap evaluation.

Code and package are available via BowDet [8].

6. AI USAGE STATEMENT

Generative AI tools (Claude Sonnet 5 and ChatGPT 5.5) were used to assist with paraphrasing, translation, and language editing during the preparation of this paper. All AI-assisted content was reviewed, revised, and verified by the authors, who take full responsibility for the submitted manuscript.

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